

FOSFORUMS 2026

Open Science

A Social Duty and its Consequences for Innovation and
Reproducibility

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About Me

SPEAKER

Carlos Henrique Craveiro Aquino Veras

- Computer Engineer from EESC/ICMC - USP.
- Master's Student in Mechanical Engineering – Aeronautics at EESC - USP.
- A science enthusiast since I can remember and an enthusiast of the FOSS movement since 2021.
- I am currently researching the operation of omnidirectional drones at offshore platforms and other interesting topics.

```
printf("Hello World!\n");
```



A decorative graphic consisting of two blue dots of different sizes, with the larger dot positioned above and to the left of the smaller dot.

1

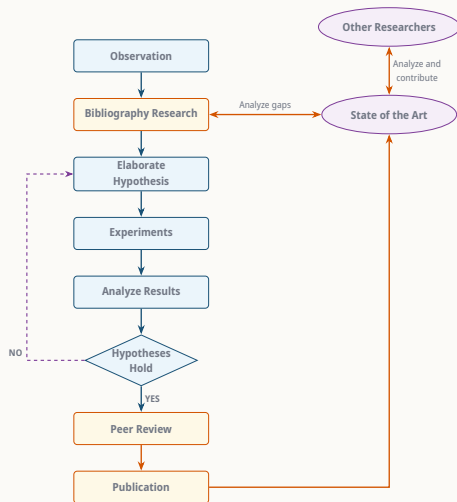
Introducing some concepts

1 Review of the Scientific Method

- **Bibliography research:** Search the state of the art for what has already been done.
- **Peer Review:** Review of the study's merit, methods, and results by external experts.
- **Publication:** Contribute the knowledge gained to the state-of-the-art knowledge base, enabling others to do the same.

Question

What happens when some points of a study, such as **methods, data, algorithms**, etc, are **unclear, missing** or **not made public**?



1 Answer: Poor Quality Science

Lack of reproducibility

Others have difficulty reproducing or even understanding the published study.

Difficulty extending the work

It gets difficult for others to take the study and extend it to other cases, or even use it for **fair comparison** (in the same conditions) in **their own studies**.

Non-falsifiable studies

The omission of certain important aspects from the study may render it unfalsifiable, given the **difficulty of verifying its claims**.

Less citations

The **lack of information** necessary for a study to be better **understood and replicated** can lead to it being less cited by the community.

1 Who Funds Scientific Research?

THE FINANCIAL BACKBONE

Historically, science is heavily supported by public resources, the State, or more specifically the **taxpayer**. Although the examples are from Brazil, the same situation follows on the rest of the world.

Public Agencies

Direct funding from the State.

FINEP CNPq CAPES
FAPESP FAPERJ FAPCE

State & Subsidies

State-owned companies and subsidized private R&D.

Petrobras (CENPES) Embrapa
El Dorado Von Braun

Public Universities

Research Centers funded by Federal and Regional states government.

USP Unicamp UNESP
UFRJ UFC UFSCar

Isn't it fair and ethical that the public have direct and access to the science they fund, especially since it also improves the quality of research?



2

**Yes! So let's talk about
Open Science**

Definition of *open scientific knowledge* according to UNESCO

The unrestricted access to publications, data, metadata, educational resources, software, source code, and hardware made available in the public domain or under open licenses, enabling access, reuse, adaptation, and distribution under specific conditions, free of charge and immediately, to all actors, regardless of their location or individual characteristics.

THE 3 CORE ELEMENTS

- **Open Access:** Publications & articles.
- **Open Data:** Raw data, models, specs.
- **Open Source/Processes:** Software, algorithms, and methodologies.

A PRAGMATIC GOAL

Accelerate scientific, technological, economic, and social progress through active **collaboration and knowledge reuse**.

2 Core Element 1: Open Access

What does it mean?

The **free**, **immediate**, and **unrestricted** online access to scholarly research, allowing anyone to read, download, use, and reuse the work for any lawful purpose provided the authors receive **proper credit**.

Traditional vs. Open

Paywalled Science

Limited Reach

Open Access

Global Impact

OPEN ACCESS HISTORY

More than 40 years of history, but four key events were:

- **Santa Fe (1999):** Interoperability and metadata protocol (OAI-PMH).
- **Budapest (2002):** Coined and defined "Open Access".
- **Bethesda (2003):** Defined open publishing and user rights.
- **Berlin (2003):** Committed global research institutions.

BEYOND JUST PUBLISHING DATA

Simply "dumping" data online is not enough. For data to be truly open and useful to the community, it must follow the **FAIR** principles:

F - Findable

Data must have rich metadata and a unique, persistent identifier (like a DOI) so both humans and computers can easily find it.

I - Interoperable

Data should be formatted in ways that allow it to be integrated with other datasets and analyzed by different software.

A - Accessible

Once found, users need to know how to access it, usually through standard, open, and universally implementable protocols.

R - Reusable

Data must be well-documented, clear in its context, and released with a highly accessible data usage license.

THE PROBLEM WITH "BLACK BOXES"

Proprietary Tools Harm Science

Closed software (e.g., MATLAB toolboxes) or other proprietary solutions create severe barriers:

- **Access Barriers:** Replicating results requires buying expensive licenses or exact equipment, locking out researchers or pushing them toward piracy.
- **Hidden Mechanics:** If a method fails on a different setup, you cannot investigate the root cause without access to the source code or schematics.

THE OPEN SOLUTION

FOSS & Open Hardware

Sharing actual code, algorithms, and hardware blueprints ensures:

- **Deep Investigation:** Full transparency to debug and understand exactly *how* and *why* a technique works.
- **Ease of Customization:** Freedom to edit intrinsic features to adapt the tool to your specific needs.
- **Community Ecosystem:** Contributions are easily assimilated, enriching available tools and fostering true collaboration.

3

Benefits of Open Science

3 For Scienc and Researchers

FOR THE SCIENTIFIC FIELD

- **Robustness:** Enables independent validation and true reproducibility.
- **Innovation:** Fosters multidisciplinary collaboration and the creation of new tools.
- **Democratization:** Reduces global inequalities by lowering access barriers to knowledge.
- **Preservation:** Guarantees long-term data integrity in public repositories.

FOR THE RESEARCHER

- **Visibility:** Expands research networks and boosts citations.
- **Resource Economy:** Reusing existing data saves time and precious funding.
[Human Genome Project](#) [Data Amazon \(USP\)](#)
- **New Analyses:** Crossing different datasets generates unique insights.
- **Integrity:** Enhances personal accountability and academic reputation.

Access to Knowledge

Open access to articles and data directly benefits **education, innovation, and social development.**

Institutional Transparency

Increases the **visibility of universities** and research centers, helps to **demonstrates their social importance** and impact.

PUBLIC ENGAGEMENT

It fosters an understanding of **how science works**, its methods and methodologies, **empowering scientific communication** and dissemination, hence its importance.

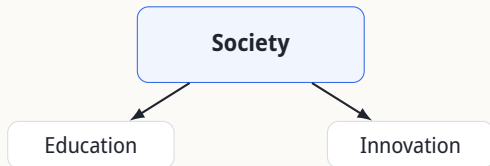




Figure 1: The Demon-Haunted World (Brazilian edition cover).

Direct access **fosters public respect** while **deepening the understanding of scientific methods**, tools, and limitations. Crucial for informed citizenship in a democratic society.

We've arranged a global civilization in which most crucial elements profoundly depend on science and technology. We have also arranged things so that nobody understands science and technology. This is a prescription for disaster. We might get away with it for a while, but sooner or later this combustible mixture of ignorance and power is going to blow up in our faces.

— Carl Sagan

(The Demon-Haunted World)

4

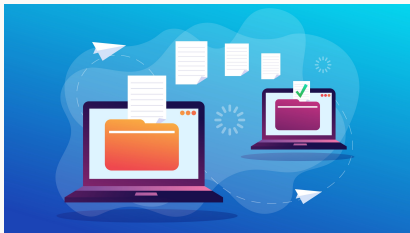
Guides for practicing Open Science



1. OPEN ACCESS PUBLISHING

Publish your work in **Open Access (OA) journals**.

- If full OA is not possible, utilize **pre-print servers** (like arXiv) to share early versions.
- *Note: Pre-prints are not peer-reviewed yet, but ensure immediate public access.*



2. SHARE AUXILIARY MATERIALS

Don't hide the "messy" but crucial details that don't fit in the main paper. Share:

- Complete mathematical deductions.
- In-depth descriptions of methods and procedures.
- Extended datasets and raw outputs.

The Main Point

Make your work easy to understand, even for professionals from other fields, and also easy to reproduce.

3. OPEN TOOLS & LICENSING

Prioritize FOSS solutions for your research. If you develop new ones, release them openly!

- **Least Alternative for Hardware:** If you really cannot make it open, **at least patent it**. Patents require a detailed description that inherently allows for future reproduction of the idea.



4. OPEN DATA (FAIR)

Publish your datasets following the **FAIR** principles.

Watch out for LGPD / GDPR!

Anonymization is key. Be careful when opening data involving human subjects. Always follow:

- LGPD (General Data Protection Law).
- Ethical and bioethical guidelines.
- Common sense to protect privacy.

4 Open Data & Privacy: The LGPD

BRAZILIAN GENERAL DATA PROTECTION LAW (LAW 13,709/2018)

For Academics: Compliance Basics

- **Informed Consent (Art. 8):** Data collection requires clear, unambiguous consent for specific research purposes.
- **Anonymized Data (Art. 12):** Truly anonymized data is **not** considered personal data under the law. The legal foundation that makes sharing Open Data possible.
- **Security (Art. 46):** Researchers must adopt strict technical measures to protect raw data from leaks before it is anonymized and published.

For the Public: Peace of Mind

Consenting **does not** mean giving up your privacy.

The Law states that shared data must consist strictly of **scientific variables**, ensuring the candidate's privacy. All names, IDs, and personal traces are stripped away before publication to guarantee that data cannot be used for discriminatory purposes.





5

The Challenges of Open Science

The Main Issue

Doing Open Science demands more effort, time, and resources than traditional “closed” science, simply because ensuring **true reproducibility is hard**, and sometimes **expensive**.

FINANCIAL AND TIME COSTS

- Article Processing Charges (APCs) for **Open Access journals** are often very **expensive**.
- Costs associated with **long-term data storage** and repository maintenance.
- Cost associated with ensuring **readability, scalability, and documentation**, as well as the time spent contributing to projects and managing a community.

TECHNICAL & HUMAN RESOURCES

- Requires specific knowledge in **data stewardship**.
- Writing and contributing to maintainable/open-source software/hardware projects demands specific skills and knowledge.
- Needs specialized personnel for **documentation, project and community management**.

GOOD NEWS FOR BRAZILIAN RESEARCHERS

Recent agreements by CAPES are helping to waive or reduce Article Processing Charges (APCs) for Brazilian authors in major publishers.

News Highlight from FAPESP Magazine - Feb 2026 - CC-BY-NC-ND. By: Sarah Schmidt

ACESSO ABERTO

Capes amplia acordos com editoras e autores brasileiros poderão publicar, sem custo, em milhares de periódicos

Novos convênios de leitura e publicação com Springer Nature, Elsevier e ACM incluem principalmente revistas de modelo híbrido

OTHER FORM OF INSTITUTIONAL SUPPORT

Some big public research institutions host their own open data "repos", such as Unicamp with its Digital Educational Repository (REDU), thus absorbing the financial cost to host the data for long periods of time.

Unicamp's Open Data Repository (REDU)



Metrics

74,478 Downloads

Contact Share

REDU é a plataforma oficial da Unicamp para o depósito, preservação, compartilhamento, reutilização e reprodutibilidade dos dados de pesquisa gerados na Universidade. Para mais informações, acesse o link contendo [o tutorial](#), [as perguntas frequentes \(FAQ\)](#) e [os canais de contato](#).



Artes e Humanas



Biomédicas



Ciências Aplicadas



Exatas



5 The Role of the Research Software Engineer (RSE)

WHO IS THE RSE?

A professional who combines a deep understanding of **research** with reasonable **software engineering** skills and knowledge.

Between Two Worlds

As we saw, technical challenges (data stewardship, code documentation, community building) are major barriers to Open Science. So, we need someone who **understands both worlds** to solve this problem.

WHY ARE THEY ESSENTIAL?

The RSE directly tackles the technical challenges of Open Science by:

- Building robust, reusable, and open research software.
- Ensuring code and data follow FAIR principles.
- Writing high-quality documentation that allows the community to actually use the tools.
- Taking the burden of “software maintenance” off the shoulders of traditional scientists.

5 One case of Personal Experience

FROM A LAB WHERE I DID A RESEARCH INTERNSHIP ABROAD

The principal investigator is interested in making the ecosystem open, but although the code is available, the documentation is not yet good enough for an outsider to reproduce and contribute to the environment without full assistance from internal staff. **Lacks a lot of crucial information.**

Dragon Lab - The University of Tokyo - Project Aerial Robot Nerve

The screenshot shows the GitHub interface for the repository 'aerial_robot_nerve' by user 'kitagawa444'. The repository is public and has 1 watch, 3 forks, and 2 stars. The main branch is 'master'. The repository description is 'Repository of aerial robot firmware (e.g. Spinal, Neuron)'. The file list includes 'spinal', 'spinal_msgs', '.gitignore', and 'README.md'. The 'spinal' directory has a recent merge pull request. The 'spinal_msgs' directory has a commit from 2 months ago. The '.gitignore' file was modified 7 months ago. The 'README.md' file was reformulated 2 months ago.

File	Commit Message	Time Ago
spinal	Merge pull request #14 from kitagawa444/PR/fix/build-error-micro-ros	10 hours ago
spinal_msgs	[Spinal][thruster] Report PWM control mode	2 months ago
.gitignore	[Git] Modified gitignore	7 months ago
README.md	Reformulate ReadMe and fix instructions	2 months ago

6

References



6 Disclaimer & Acknowledgments

AUTHOR'S RESPONSIBILITY

I take full responsibility for the content, opinions, and technical claims presented in this talk. All scientific perspectives are my own.

AI Assistance Note

Generative AI tools were used to assist in the preparation of this presentation:

- Pre-formatting the Beamer slides.
- Textual review and translation.
- Generation of the TikZ flowchart.

- Max Planck Society (2003).
Berlin Declaration on Open Access to Knowledge in the Sciences and Humanities.
<https://openaccess.mpg.de/Berlin-Declaration>
- Coursera & University of São Paulo.
Boas Práticas Científicas (Good Scientific Practices Course).
<https://www.coursera.org/learn/boas-praticas-cientificas>
- FAPESP.
FAPESP Open Science.
<https://www.fapesp.br/openscience/>
- MDPI Blog (2026).
4 Defining Declarations in the History of Open Access.
<https://blog.mdpi.com/2026/08/04/4-defining-declarations-in-the-history-of-open-access/>
- National Academies of Sciences, Engineering, and Medicine (2018).
Open Science by Design: Realizing a Vision for 21st Century Research.
Washington, DC: The National Academies Press. <https://doi.org/10.17226/25116>
- Presidência da República, Brazil (2018).
Lei nº 13.709: Lei Geral de Proteção de Dados Pessoais (LGPD).
https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/lei/l13709.htm

- Revista Pesquisa FAPESP.
CAPES amplia acordos com editoras e autores brasileiros poderão publicar sem custo.
<https://revistapesquisa.fapesp.br/capes-amplia-acordos-com-editoras-e-autores-brasileiros-poderao-publicar-sem-custo-em-milhares-de-per>
- Sagan, Carl (2006).
O mundo assombrado pelos demônios: a ciência vista como uma vela no escuro.
São Paulo: Companhia de Bolso. ISBN 978-85-359-0834-3.
- SciELO in Perspective (2013).
Evolução do Acesso Aberto: breve histórico.
<https://blog.scielo.org/blog/2013/10/21/evolucao-do-acesso-aberto-breve-historico/>
- Sociedade Brasileira para o Progresso da Ciência (SBPC) (2025).
Ciência Aberta: O caminho para a democratização do conhecimento e a inovação colaborativa.
Ciência & Cultura, 77(1). <http://hdl.handle.net/20.500.11832/10009>
- Society of Research Software Engineering.
Society of RSE Official Website.
<https://society-rse.org/>

- TaxonWorks (2025).
What's a Research Software Engineer (RSE)? [Video]. YouTube.
<https://youtu.be/PQuSBQSjlBc>
- UNESCO.
Open Access to Scientific Information.
<https://www.unesco.org/en/open-access>
- UNICAMP.
REDU - Repositório de Dados de Pesquisa da Unicamp.
<https://redu.unicamp.br/>
- Universidade de São Paulo.
Declaração USP de Apoio à Ciência Aberta.
<https://cienciaaberta.usp.br/declaracao-usp-de-apoio-a-ciencia-aberta/>
- UT Dragon Lab. *Aerial Robot Nerve (Source Code Repository).*
https://github.com/ut-dragon-lab/aerial_robot_nerve
- Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. (2016).
The FAIR Guiding Principles for scientific data management and stewardship.
Sci Data 3, 160018. <https://doi.org/10.1038/sdata.2016.18>

That's All Folks!

Questions and Discussions are now welcome.

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https://carloscraveiro.github.io/personal_blog/

